

ORDER AND ORDER ITEMS MANAGEMENT

Project Title: **IKEA – E-Commerce Database Management System**

1.Database design

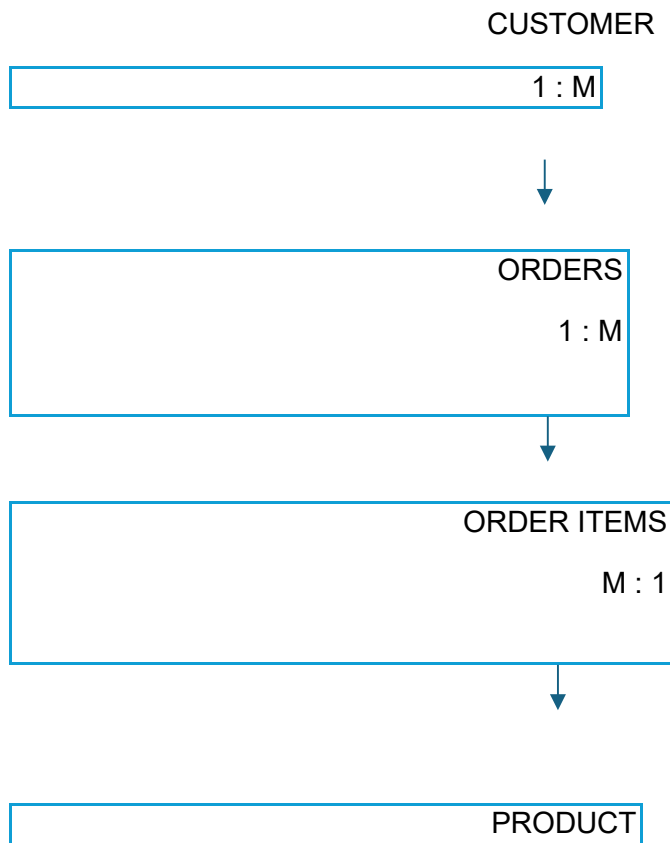
1.1 Orders Table

Attribute	Data Type	Key	Description
Order_ID	INT	PK	Unique order ID
Customer_ID	INT	FK	References Customer
Order_Date	DATE	-	Order date
Total_Amount	DECIMAL(10,2)	-	Total amount
Order_Status	VARCHAR(30)	-	Placed/Shipped/Delivered
Shipping_Address	TEXT	-	Shipping address
Delivery_Date	DATE	-	Delivery date

1.2 Orders_items Table

Attribute	Data Type	Key	Description
Order_Item_ID	INT	PK	Unique order item ID
Order_ID	INT	FK	References Orders
Product_ID	INT	FK	References Product
Quantity	INT	-	Quantity
Unit_Price	DECIMAL(10,2)	-	Unit price
Discount	DECIMAL(5,2)	-	Discount
Total_Price	DECIMAL(10,2)	-	Total price

2. Table Relationships



3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Orders (  
2     Order_ID INT PRIMARY KEY,  
3     Customer_ID INT,  
4     Order_Date DATE,  
5     Total_Amount DECIMAL(10,2),  
6     Order_Status VARCHAR2(30),  
7     Shipping_Address VARCHAR2(255),  
8     Delivery_Date DATE,  
9     CONSTRAINT fk_orders_customer  
10         FOREIGN KEY (Customer_ID)  
11         REFERENCES Customer(Customer_ID)  
12 );
```

Table created.

3.2 Create Order_Details

```
SQL> CREATE TABLE Order_Items (  
2     Order_Item_ID INT PRIMARY KEY,  
3     Order_ID INT,  
4     Product_ID INT,  
5     Quantity INT,  
6     Unit_Price DECIMAL(10,2),  
7     Discount DECIMAL(5,2),  
8     Total_Price DECIMAL(10,2),  
9     CONSTRAINT fk_orderitems_order  
10         FOREIGN KEY (Order_ID)  
11         REFERENCES Orders(Order_ID),  
12     CONSTRAINT fk_orderitems_product  
13         FOREIGN KEY (Product_ID)  
14         REFERENCES Product(Product_ID)  
15 );
```

Table created.

4. Order Data Insertion

```
SQL> INSERT INTO Orders
  2 VALUES (501, 101, TO_DATE('10-SEP-2026', 'DD-MON-YYYY'),
  3          1231.12, 'Delivered', '12 Anna Nagar, Chennai',
  4          TO_DATE('14-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (502, 102, TO_DATE('11-SEP-2026', 'DD-MON-YYYY'),
  3          3398.30, 'Shipped', 'Chennai',
  4          TO_DATE('16-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (503, 103, TO_DATE('12-SEP-2026', 'DD-MON-YYYY'),
  3          2799.20, 'Delivered', 'Coimbatore',
  4          TO_DATE('17-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (504, 104, TO_DATE('13-SEP-2026', 'DD-MON-YYYY'),
  3          5699.05, 'Shipped', 'Bangalore',
  4          TO_DATE('18-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (505, 105, TO_DATE('14-SEP-2026', 'DD-MON-YYYY'),
  3          1619.10, 'Delivered', 'Coimbatore',
  4          TO_DATE('19-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (506, 106, TO_DATE('15-SEP-2026', 'DD-MON-YYYY'),
  3          2462.24, 'Placed', 'Madurai',
  4          NULL);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

```
SQL> SELECT * FROM Orders;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
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SHIPPING_ADDRESS				
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DELIVERY_				
-----------	--	--	--	--

501	101	10-SEP-26	1231.12	Delivered
12 Anna Nagar, Chennai				
14-SEP-26				

502	102	11-SEP-26	3398.3	Shipped
Chennai				
16-SEP-26				

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
----------	-------------	-----------	--------------	--------------

SHIPPING_ADDRESS				
------------------	--	--	--	--

DELIVERY_				
-----------	--	--	--	--

503	103	12-SEP-26	2799.2	Delivered
Coimbatore				
17-SEP-26				

504	104	13-SEP-26	5699.05	Shipped
Bangalore				

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
----------	-------------	-----------	--------------	--------------

SHIPPING_ADDRESS				
------------------	--	--	--	--

DELIVERY_				
-----------	--	--	--	--

18-SEP-26				
-----------	--	--	--	--

505	105	14-SEP-26	1619.1	Delivered
Coimbatore				
19-SEP-26				

506	106	15-SEP-26	2462.24	Placed
-----	-----	-----------	---------	--------

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
----------	-------------	-----------	--------------	--------------

SHIPPING_ADDRESS				
------------------	--	--	--	--

DELIVERY_				
-----------	--	--	--	--

Madurai				
---------	--	--	--	--

5. Order_items Data Insertion

```

SQL> INSERT INTO Order_Items
  2 VALUES (601, 501, 201, 1, 1399.00, 12.00, 1231.12);

1 row created.

SQL>
SQL> INSERT INTO Order_Items
  2 VALUES (602, 502, 202, 2, 1999.00, 15.00, 3398.30);

1 row created.

SQL>
SQL> INSERT INTO Order_Items
  2 VALUES (603, 503, 203, 1, 3499.00, 20.00, 2799.20);

1 row created.

SQL>
SQL> INSERT INTO Order_Items
  2 VALUES (604, 504, 204, 1, 5999.00, 5.00, 5699.05);

1 row created.

SQL>
SQL> INSERT INTO Order_Items
  2 VALUES (605, 505, 205, 1, 1799.00, 10.00, 1619.10);

1 row created.

SQL>
SQL> INSERT INTO Order_Items
  2 VALUES (606, 506, 201, 2, 1399.00, 12.00, 2462.24);

1 row created.

SQL>

```

```

SQL> SELECT * FROM Order_Items;

```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	DISCOUNT	TOTAL_PRICE
601	501	201	1	1399	12	1231.12
602	502	202	2	1999	15	3398.3
603	503	203	1	3499	20	2799.2
604	504	204	1	5999	5	5699.05
605	505	205	1	1799	10	1619.1
606	506	201	2	1399	12	2462.24

```

6 rows selected.

```

6. Order Modification Operations

6.1 Modify Order Amount:

```

SQL> UPDATE Orders
  2 SET Total_Amount = 1300.00
  3 WHERE Order_ID = 501;

1 row updated.

SQL>
SQL> SELECT * FROM Orders
  2 WHERE Order_ID = 501;

ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
          501          101 10-SEP-26          1300 Delivered
12 Anna Nagar, Chennai
14-SEP-26

```

6.2 Modify Order Date:

```

SQL> UPDATE Orders
  2 SET Order_Date = TO_DATE('16-SEP-2026','DD-MON-YYYY')
  3 WHERE Order_ID = 502;

1 row updated.

SQL>
SQL> SELECT * FROM Orders
  2 WHERE Order_ID = 502;

ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
          502          102 16-SEP-26          3398.3 Shipped
Chennai
16-SEP-26

```

7. Order_Items Modification

7.1 Modify Order Item

```

SQL> UPDATE Order_Items
  2 SET Quantity = 2,
  3     Total_Price = 2462.24
  4 WHERE Order_Item_ID = 601;

1 row updated.

SQL>
SQL> SELECT * FROM Order_Items
  2 WHERE Order_Item_ID = 601;

```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	DISCOUNT	TOTAL_PRICE
601	501	201	2	1399	12	2462.24

7.2 Order_Items Modification

```

SQL> UPDATE Order_Items
  2 SET Quantity = 2,
  3     Total_Price = 2462.24
  4 WHERE Order_Item_ID = 601;

1 row updated.

SQL>
SQL> SELECT * FROM Order_Items
  2 WHERE Order_Item_ID = 601;

```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	DISCOUNT	TOTAL_PRICE
601	501	201	2	1399	12	2462.24

```

SQL> UPDATE Order_Items
  2 SET Quantity = 2,
  3     Total_Price = 2462.24
  4 WHERE Order_Item_ID = 601;

1 row updated.

SQL>
SQL> SELECT * FROM Order_Items
  2 WHERE Order_Item_ID = 601;

```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	DISCOUNT	TOTAL_PRICE
601	501	201	2	1399	12	2462.24

8. Customer Order History Report

```

SQL> SELECT o.Order_ID,
2         o.Customer_ID,
3         o.Order_Date,
4         o.Total_Amount,
5         o.Order_Status,
6         oi.Product_ID,
7         oi.Quantity,
8         oi.Unit_Price,
9         oi.Discount,
10        oi.Total_Price
11 FROM Orders o
12 JOIN Order_Items oi
13 ON o.Order_ID = oi.Order_ID
14 ORDER BY o.Order_ID;

```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
501	101	10-SEP-26	1300	Delivered
201	2	1399	12	2462.24
502	102	16-SEP-26	3398.3	Shipped
202	2	1999	15	3398.3
503	103	12-SEP-26	2799.2	Delivered
203	1	3499	20	2799.2

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
504	104	13-SEP-26	5699.05	Shipped
204	1	5999	5	5699.05
505	105	14-SEP-26	1619.1	Delivered
205	1	1799	10	1619.1
506	106	15-SEP-26	2462.24	Placed
201	2	1399	12	2462.24

9. Customer-wise Order Summary

```

SQL> SELECT c.Customer_ID,
2         c.First_Name,
3         c.Last_Name,
4         COUNT(o.Order_ID) AS Total_Orders,
5         SUM(o.Total_Amount) AS Total_Amount
6 FROM Customer c
7 JOIN Orders o
8 ON c.Customer_ID = o.Customer_ID
9 GROUP BY c.Customer_ID,
10         c.First_Name,
11         c.Last_Name
12 ORDER BY c.Customer_ID;

```

CUSTOMER_ID	FIRST_NAME		
LAST_NAME		TOTAL_ORDERS	TOTAL_AMOUNT
101 Rahul	Sharma	1	1300
102 Priya	Ravi	1	3398.3
103 Karthik	Kumar	1	2799.2

CUSTOMER_ID	FIRST_NAME		
LAST_NAME		TOTAL_ORDERS	TOTAL_AMOUNT
104 Anitha	Suresh	1	5699.05
105 Arun	Prakash	1	1619.1
106 Divya	Lakshmi	1	2462.24

6 rows selected.

